

MAKE UP EXAMINATIONS – APRIL 2025

Program	: B.E :- Common to all Programs	Semester	: V
Course Name	: Research Methodology and Intellectual Property Rights	Max. Marks	: 100
Course Code	: AL58	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.
- Use of Statistical tables is permitted.

UNIT - I

- Explain the different types of research. CO1 (10)
 - Highlight the significance and objectives of research. CO1 (10)
- How does knowledge flow happen through citations? Explain with a flowchart. CO1 (08)
 - How do the title of research and the keywords used in the research impact citations? Explain. CO1 (07)
 - Why are critical and creative reading important in research? Explain. CO1 (05)

UNIT - II

- What is the need for research design? Explain the features of a good design. CO2 (10)
 - Describe Latin Square Design (L.S. Design) in detail with examples. CO2 (10)
- Write short notes on the following: CO2 (10)
 - Dependent and independent variables
 - Extraneous variable
 - Experimental and control groups.
 - Explain randomized block design along with its advantages and disadvantages. CO2 (10)

UNIT - III

- How are the interview method and questionnaires different in data collection? Give valid illustrations. CO3 (08)
 - Define confidence level and significance level. CO3 (04)
 - A teacher believes that 85% of students in the class will want to go on a field trip to the local zoo. She performs a hypothesis test to determine if the percentage is the same or different from 85%. The teacher samples 50 students and 39 reply that they would want to go to the zoo. Conduct hypothesis test, using 5% level of significance. CO3 (08)
- Define hypothesis. Illustrate the procedure for hypothesis testing. CO3 (10)
 - The average adult male height in a certain country is 170 cm. We suspect that the men in a certain city in that country might have a different average height due to some environmental factors. We pick a random sample of size 9 from the adult males in the city and obtain the following values for their heights (in cm): 176.2, 157.9, 160.1, 180.9, 165.1, 167.2, 162.9, 155.7, 166.2. Test whether average male height can be 170cm. Consider significance level to be 5%. CO3 (10)

UNIT- IV

7. a) List all the issues and features covered in TRIPS agreement. CO4 (10)
b) What are Patent Infringements? Explain its two categories of Infringements. CO4 (10)
8. a) Define the term patent and what are the conditions that must be met for obtaining patent protection? CO4 (10)
b) Explain the following steps involved in registration of a patent. CO4 (10)
(i) Prior Art Search (ii) Choice of Application to be Filed (iii) Pre-grant Opposition.

UNIT - V

9. a) Explain the essential requirements of a registrable design. CO5 (10)
b) How do you register a design in India? Explain. CO5 (10)
10. a) Identify the rights that are conferred by registration of trademarks. CO5 (06)
b) Compare and contrast copyright and patents. CO5 (06)
c) What are infringements? what are the remedies against infringements of copyrights? CO5 (08)
